Harsh Raj

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AI/ML Engineer

B.Tech. in Mechatronics and Automation Engineering — Indian Institute of Information Technology, Bhagalpur

EDUCATION

Indian Institute of Information Technology, Bhagalpur *2024–2028*
B.Tech. in Mechatronics and Automation Engineering **CGPA: 7.81/10**
Minor in Computer Science and Engineering (CSE), IIIT Bhagalpur
New Holy Ganges Public School, Khagaria, Bihar Class XII: 84%

TECHNICAL SKILLS

Languages: Python, C++

Data Science: Data preprocessing, exploratory data analysis, statistical analysis, feature engineering, data visualization

Machine Learning: Scikit-learn, supervised and unsupervised learning, classification and regression, ensemble methods, model evaluation, cross-validation, hyperparameter optimization

Deep Learning: PyTorch, TensorFlow/Keras, neural networks, CNNs, transfer learning

NLP & Computer Vision: Text processing, embeddings, Transformers, OCR, image processing

Generative & Agentic AI: LLMs, LangChain, LangGraph, RAG, prompt engineering, tool calling, multi-agent systems, LLM orchestration, Hugging Face Transformers, SentenceTransformers

Backend, Deployment & Tools: FastAPI, Streamlit, Pydantic, REST APIs, Supabase, AWS EC2, Hugging Face Spaces, Git, GitHub, BeautifulSoup, lxml

CS Fundamentals: OOP, DBMS, Operating Systems, Data Structures and Algorithms

PROJECTS

CogniSynth: Multi-Agent Research System *Jun 2026–Present*
Python, LangGraph, Gemini, Tavily, FastAPI, Streamlit GitHub | Live Demo

- Built a multi-agent research workflow that decomposes a topic into research tasks and coordinates web search, page extraction, claim analysis, verification, synthesis, and report generation.
- Implemented evidence-linked claims, source-quality scoring, contradiction detection, and persistent local semantic memory to make findings more traceable and easier to review.
- Developed a Streamlit dashboard with FastAPI endpoints and server-sent events (SSE), enabling users to submit research requests and follow workflow progress as results are produced.

Brand AI-Readiness Auditing Engine *Aug 2026–Present*
Python, Web Crawling, HTML Parsing, JSON, FastAPI, React GitHub | Live Demo

- Developed an audit engine that assesses crawler access, HTML quotability, structured brand/entity information, content freshness, user-action pathways, and barriers caused by non-text content.
- Added checks for crawler directives, sitemap and metadata signals, structured information, and content discoverability to identify issues that may limit website visibility to AI-driven systems.
- Built a bounded, robots-aware crawling pipeline that returns structured JSON findings with supporting evidence, severity-ranked recommendations, and verification steps to help reduce false positives.

MediReach: AI-Enabled Healthcare Platform *2026*
Team Project | LLM Services, OCR, Document Processing, FastAPI GitHub

- Built an LLM-driven patient-intake workflow to collect and structure patient information, with a document pipeline to extract data from uploaded medical records.
- Developed AI-assisted report generation and protocol-oriented review; an LLM reviewer flags contradictions and unsupported claims in generated reports.
- Integrated AI services with FastAPI and contributed to SQLite persistence, API-key authentication, configurable multilingual support, medicine-label OCR, and voice-assistance workflows; team secured second place at the SIH '26 internal hackathon.

ACHIEVEMENTS AND CERTIFICATION

View Certificates

- **3rd Place**, AI-Verse '26 Machine Learning Challenge, IIIT Bhagalpur (2026); **2nd Place**, SIH '26 Internal Hackathon (2026).
- Solved **200+ DSA problems**; profile: Codolio.
- **Introduction to Large Language Models**, IBM SkillsBuild, Jan 2026.

LEADERSHIP

Project Manager, Robotics and Electronics Club, IIIT Bhagalpur *2024–2025*

- Coordinated team members, supported task planning, and oversaw the execution of robotics projects within club activities.
- Organized technical workshops and guided junior students during robotics project work and technical learning activities.